ActInf Textbook Group ~ Cohort 8 ~ Session 2 (Onboarding and Chapter 1) ~ 2/27/2025

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[Music] yeah go for it hi everyone welcome to the active inference Institute the textbook reading group which we hold uh twice every Thursday and this is the latter group cohort 8 reading chapter one uh second week of that and so we're being introduced to active inference um this is with chapter one it's uh one of the shorter chapters of the book of course it's just opening things up so it doesn't jump too quickly to say any complex formalisms or the definition of expected free energy just yet we're largely just introduced to sort of uh the broader Outlook of active inference is a normative framework for approaching uh Behavior action inferential processes uh with a a foundational sort of neural process Theory and and functional that is free energy and free energy minimization so uh quickly we move to looking at living organisms or another more uh kind of abstract notion of that which we now hear all the time uh for those who work in Ai and Tech uh agents as a sort of a general notion of what could be an organism a system of interest that exhibits Behavior Uh or or otherwise um agents Tak in sensory observ ations and elicit actions to either one realize their desired outcomes or goals or two to make further sense of their world by gaining information this General overview leads to the development of adaptive behaviors or strategies agents use for navigating their environment and this process plays out in a recursive action perception cycle where observations inform agents decision-making regarding the next action they should take while we may inure future often view this as a kind of discrete setting where there are multiple time steps or Cycles where you could simulate an agent in reality uh we tend to view these more as continuous processes as we're aware that much of active inference actually derives from not just the conceptual work and Mathematics but the direct empirical work uh friston and others have done whenever it comes to uh working with neuroimaging data and dynamic causal modeling other forms of looking at continuous processes over time and linking them with observed Behavior so these inference processes or action perception Cycles can occur at various levels of complexity and time scales from the simple and rigid M by moment strategies of nutrient seeking bacteria to the more complex and flexible strategies of humans engaging in planning to realize their goals in the future potentially all the way up to Broad cultural and developmental learning in collectives societies so um they they give us an interesting sort of way of viewing active inference and perhaps why or why they might justify uh introducing it as a fully normative framework and so they frame it as follows uh active inferences approach is um uh on the side of the scruffies while scruffies lay emphasis on the ID pardon me the neats while scruffies lay emphasis on the idiosyncrasies and specific ities of various phenomena of Interest such as different biological adaptations neuronal Dynamics or cognitive mechanisms each deserving separate bottomup research uh from empirical data nits aspired to a set of first principles for explaining and relating these phenomena in a more top-down fashion while still engaging an empirical analysis for validation active inference also identifies with needs as it lays a normative framework for understanding the brain and mind the biolog and cognitive implications alongside model design testing for empirical analysis this is frequently noticed if anyone uh begins to look at Journal articles or other Publications from active inference where sure there are many times where people are modeling and remaining within an artificial environment they may even be intending to design something that could be used in the realm of Robotics or uh artificial intelligence that said there are many other Behavioral studies done such as the realm of computational Psychiatry or classic uh uh lab studies with with Myer rodents where there is indeed forms of model fitting and revision of the model based upon the actual empirical data that is collected shouldn't leave out also other continuous State space and hybrid State space um examples including studying things like is secades and um neuroimaging data fluctuation and electrography and so on and we do have a question so yeah John go for it uh yeah this confused me every time I read it why why is it called a normative framework it seems to associate normative framework with first principles I've never heard an association like that before I hoping someone could clarify that's interesting yeah is that we could open up to discussion if anyone has any thought on that um otherwise I can go for it yeah is this a we could think of it as sort of an expression of natural law just a this is a physical Dynamic principle yeah go for so um I mean as far as normative in the sense of establishing a norm my take there on on this specific point is that this is in fact providing like a an attempt at a holistic framework where there are a variety of Concepts that are introduced not necessarily uh introduced aresh as active inference draws from a variety of disciplines um but nonetheless it it does attempt to build up something like potentially totalizing picture of how we could View Behavior Uh inference uh and anything from biological to other physical uh and psychological phenomena so in that sense it is drawing from first principles because it needs to establish certain principles of how to achieve that framework and maintain it and like on which it the framework relies and then normative I I I see those as rather part partial um so so that's been my take on it so if I switched normative for holistic I couldn't go too far wrong do you think I mean I would say so yeah just curious though because this might be helpful for others too um I mean what what is your because I'm looking at a quick definition of normative just to kind of lay some some connection here that that we can all work with establishing relating to or deriving from a standard or Norm especially Behavior so we do have like kind of an establishment of like this is how this is our Norm that we have for viewing behavior um holistic sure sure that that could work as well yeah yeah this is a really interesting question and I'm sure unless there's some British connotation or something peculiar which is I think a great starting point for discussion but it's not a declarative you know so it's it's it's all just language samples but it's like what is normative about how things actually are I mean are things not as they normally are according to like Central limit theorem in the limit of large classical things are described by Norms especially if we're not going to be interpolating um secondary functions like reward functions that has been applied even in the social like social scripts and expectations theory of mind and that's not even necessarily ACM that's just like expectations and norms and firstness and inertia and there just has to be an overwhelming element of things identifiable things being identifiably as they are so that's and then there's a sort of interpretation of like modeling practice normativity like you ought to consider multi-agent systems or but those might have those don't necessarily have axiomatic there might be very good epistemic hygiene or modeling strategies or machine learning or interpretability or anything like or performance um or model efficacy or something but I but it's a little different than and also what first Prin principles are Stephen yeah I think uh Andrew did a good job of uh probing what the issue is there are definitions of what normally normalizing is as a scaling but also uh what you're referring to is the mindset first principles the normative first principles are you know energy entropy uh principles and that's what uh the way it's used here I think it's a it's a fun another principle yeah yeah know that's really well said Stephen it's uh because it's yeah and what Daniel pointed to referring to social scripts and things like that I mean it the word normative not that I'm necessarily an momology but I mean it's been used in a variety of contexts and I personally have like this sort of General social sciences background before I I kind of took a turn into kind of the harder Sciences as well as data and empirical data um but uh so so normative has also been used in many contexts to refer as something that is culturally normative meaning it's it's almost as if you're saying oh doesn't really rely upon anything other than the fact that we do it and influence one another such that it's a social phenomena of you know that that that is normative in our society but perhaps not in other societies versus what I view active inference which is saying you know I mean I don't want to open things up and make it too abstract but it's kind of like it's as if freston friston at all here are saying that where kind of putting up this normative framework perhaps you choose perhaps you know perhaps it's a choice that you decide to kind of borrow into this framework at least I mean in our context obviously you know this is not a very authoritarian authoritative uh command rather it's just in the same way we spoke at the beginning of the session just prior that this can be a kind of toolkit uh for people you know it's your choice ultimately if you decide to agree with the entirety of the framework and i' honestly advise that you not do that if you've only read the first few pages of chapter one you should spend more time with it as you should with anything before fully buying into it but um I think it provides a pretty strong argument personally um you could maybe guess that given I'm here but um yeah so so there's a lot of nuance to to the word normative but I I think this is in some ways it's like an offering of a full framework that's been worked on for at least a couple of decades now and arguably longer if we kind of Trace the lineage there um and uh yeah it's a rather holistic view reading that question um that I uh I'll put in the chat this was just as as usual the notes are scattered so again it's just a starting point but here's another interesting uh view was um the term normative this could have been in alism or something the term normative means there is some evaluative standard against which behavior can be scored active inference is normative in the sense that perception and action are scored by free energy a quantity we will unpack through this in the next chapters actually maybe that's just from the book or a footnote but that's just one way of them even saying it so it again it's like you can always just say okay does it really matter even what we think about this one English word doesn't it matter more what it's actually going to be not just um um zooming in only on one adjective just because we wouldn't have written it that way in our view but giving norms and measurement giving a measure theory of information spaces which is more like the basing mechanics view there's a gauge and a norm and and measurements that that get but if we're not if we're not clear if we're talking about inculturation normativity or mathematical gauge and like sheath then I mean then it's just too broad to be useful it's just not a yeah I just didn't know what was intended by that word because I've never seen that word in context like that before sure sure yeah I it's it's it's been used in philosophy as well at certain points to distinguish like oh this is a a normative framework in the sense of it it is making claims about something that you necessarily have to agree with in order to actually get something out of or to like properly utilize the framework um it's not used in that way 100% of the time though um so actually a lot of the ambiguity around the word normative here yeah just uh it's quite broad could uh could go many ways but I I hope yeah so anyway I hope that the perhaps the word holistic is a a fair enough interpretation of the word um it I think it adds to it if it doesn't Encompass everything it probably doesn't Encompass everything so um does anyone else have any any thoughts or questions or anything they'd like to throw out there okay um so in reference to some of what we discussed as well at the very beginning of the session so this chapter of course remains rather brief relative to others and uh as we're just being introduced and so of course in a nice fashion they're nice enough to uh give us the outline of the book so the book is separated into two parts the first half focuses on Theory which I know many will be quite interested in as opposed to the second half uh this first half acts as a broad primer which covers active inference as a normative framework with its distinct Concepts and theories the second half will cover practice by describing examples of core active inference models with their components and examples of applications as well as information needed for Designing novel models and uh what they also don't mention involves a model analysis and potential fitting and other perspectives we can take on the modeling process itself so we're provided a conceptual road map for coming to an understanding of active active inference in the first half of this book which could be followed along a low road starting from basis theorem or a high road starting from what is known as the free free energy principle so the the low road especially it supplies the mechanics of active inference in its relation to the basian brain hypothesis whereby inference is seen to work by way of basis theorem so we have uh in basis theorem we have prior information about hidden States we have a likelihood distribution describing how they relate to observations and then we have new observations themselves as evidence which can be brought together to all compute posterior beliefs about in States then on the other hand with chapter three we'll get into the free energy principle necessity for agents to minimize this quantity as a proxy or upper bound on surprisal in order to survive and adapt to their environment whereby the tractable proxy allows for V variational inference and can certainly relate to many Notions of things like metabolism uh you know saving of resources expenditure um actually being able to accomplish physiologically neurobiologic neurobiologically or otherwise different kinds of inference processes that otherwise may not work or be very very strenuous to an organism to to accomplish if they they could actually run it in full uh as if they could activate every neuron in their brain simultaneously and then finally eventually we we'll get to chapter four uh which gets us back on the low roads Foundation and bases theorem so there will be a bit of a cross back to return to that previous material we're introduced to two template models used in active inference especially uh the these two exemplars the first being a PDP in discrete time scenarios and we'll also look at continuous time scenarios where we instead rely a bit more on on on different differential equations or tailor series approximation in chapter 5 uh that's kind of the meat of active inference as far as sort of making these connections to neurobiological processes goes we'll look a lot at uh variational message passing and uh their relationship to cortical circuitry and the sort of passing of information as well as uh being able to see how some of the concepts will be introduced to along the way such as ideas of precision um and other model components relate to neurobiology in the form of synaptic gain uh neurotransmitters and so on so um from there the chapter starts to wrap up and we see that U active inferences approach to various Concepts and con mechanisms such as perception which is changing one's mind and action is changing the world learning is similar to perception but operating at a slower time scale with longer term impact on the inference process so we have a separation separation between inference and learning in this framework then planning is a means of selecting a actions in sequences to realize future goals if I want some uh food I'm going to have to take several steps to get there thus devising my own policy which can then be compared to other policies um and then we also can break that up those policies and the way I select them in terms of exploration and exploitation um do I want to try a new restaurant do I want to try something different along the way do I want to stop and look online at uh local restaurants before even getting straight to the goal of actually like attaining nourishment uh so so it's it's it's worthwhile to consider like in an everyday sense given the sort of holism of this framework you know how can we relate it to everyday things I've just seen that as being useful for as a is a sort of learning horis uh for people who still who are still new to thinking in these these kinds of ways and um so on so let's see I think it's worthwhile to given the amount of time we have left it's worthwhile to just kind of open it up from there uh if anyone had any particular questions they want to pursue further they have a certain section in the in the textbook that particularly piqu their interest um we can always go to the questions for the chapter that have been asked historically Beyond this group so anyone feel free to recommend anything so the way this works is you try to update your beliefs first and if that's not good enough then can you act to change the world not necessarily um so but that is a good question uh because it's really important to clarify those two things so an active inference it's as if at any given moment someone could be doing both of these things simultaneously um that is uh I I could be seeing something and then I associate that with a hidden State set of hidden States or otherwise to to finding the beliefs uh that I have about you know what what goes beyond what I'm seeing like what something means and then simultaneously I can act right so if I'm sitting here I'm thirsty and then I look over and I see that my water bottle is there and I have a belief that you know I had put water in it and that it is available to me all the way down to this highly granular level that likely involves a lot of inferential processes that I am not at all explicitly thinking about because it's such this is such habitual behavior for me so I don't have to put that much effort in but if I actually thread it out word it in this way that there's a kind of perceptual in inference process going on and simultaneously I want water so I'm like kind of driven towards something so I also take this water bottle to be a kind of affordance to me that I can then you know proceed to act and I can pick it up um and I could go straight for the drink or I could actually look inside of it first to get more information about if it even has water in water in it or not prior to Jumping straight to the kind of pragmatic value of consuming the beverage um so so yeah if if that makes sense it's like these things can play out simultaneously and presumably they are all of the time so uh my body has a variety of circuitry within it where inferential processes can potentially be taking place or you know you could view the mind as is doing that many ways through different cortical layers um passing messages between them inferring different things and uh so on so did that give any clearer picture I'll add one thing that was very interesting Andrew is uh the the physics like basy and mechanics is in terms of uh simultaneous unfolding flow Dynamics kind of like a heat bath situation with unfolding flows that might be sampled from whereas especially if you're going to be writing an implementation that runs on a single core CPU there will be some sequential logic or program flow so there's actually a lot of degrees of freedom in between the analytical representations which are basically unopinionated on what order you do what actual implementations and so there's like this like UDA Loop or the test operate test exit all these different ways of thinking about well does this happen first or when does this belief updating happen it is certainly not specified like that in the general part of the theory because it's to be specified in the specific parts of the theory I think it's probably also important to remember there are hierarchical orders of organization you know from the on the physical level and subatomic Atomic up through on the biological scale Cellular Systems organism these different levels of organization have their own characteristic Dynamics and so if you're speaking with the language of the organismal level say the mind level of having preconceptions with constant updating based on activity sensory and motor activity that's a different hierarchical level than say on the uh thermodynamic chemical level where you might want to model it differently implementation might be slightly different but here's the here's here's what I I'm really interested Ed in the difference between the normative principle or the holistic framework versus the implementation uh whether it's modeling like you're saying Daniel on a single core system or uh just algorithmically like how to set up the implementation of what we're considering to be a a general holistic uh just natural law principle of how things work and and to that end if I can Circle back to the definitional clarifications we were trying to get at in the beginning uh Andrew how do you go through the meta metacognitively metacognitive uh process of like uh doing online translations of these words like the action and the inference uh when you're applying it in a specific uh case or application yeah to to be clear do you mean to say in the in the sense of like if I were actually going to sit down and engage in some kind of cognitive modeling or do you mean like just the literal translation a moment ago of me describing picking up a a glass of of water yeah e either one would work just either one yeah cool uh no that uh and that and I'll also say that the two would kind of borrow from one another um so maybe I can since it is a personal question uh which I appreciate maybe I will just I'll just go for it um let's see so what really helped me personally when engaging in like actually attempting to model something which so for anyone who is interested in modeling to the degree of they want to program themselves um a lot of active inference modeling uh there are growing numbers of resources and tools for accomplishing that I kind of starts with friston's work in the mat lab language and and building his own um SPM uh package for that and U I'll I'll I'll skip ahead a little bit and say that um to actually start developing something where you can simulate an organism or agent within its environment like actually kind of coding that up uh as a simulation then watching it play out and watching these kind of like algorithmic steps that play out I think it's highly informative like in that kind of intuitive sense of like and also active inference would say this as well right we're not just passive observers we actively engage with the world including to seek information so all I mean to say with that is by actually sitting down to do the cognitive modeling that actually filled in a lot of gaps for me not to say that someone who doesn't program could never understand active inference certainly not the case uh we've had so many people who don't know how to code and yet they have very kind of Rich backgrounds uh in in philosophy and psychology and so on right now I'm working on a project with um some folks at The Institute where one of them is a clinical therapist and so I'm kind of playing the role of the the computational psychiatrist where we're devising a model together which relies upon active inference foundational uh kind of first principles perspective on how to do this and then the clinical therapist has reported uh information from from Anonymous uh well they know them but redacted from the final study uh participants they engaged with who all of whom had uh were diagnosed at one point with PTSD so whenever I sit down to think about how to model that and of course this is very sensitive subject matter so I don't really take it lightly to just sit down and think oh yeah I can just make a person who exhibits all the characteristics of PTSD and then one sitting let alone like you know I would want to fit that model to real world phenomena for example obviously um so all that said um I guess with perception and action you know like I said earlier it tends to play out in a continuous setting in reality but if we're going to set up kind of a sequential like step by step like following the classic roles of like developing models that are not too complex while still being uh uh accurate like a kind of einsteinian you know simple as possible but no simpler sort of approach then then it becomes useful to especially in the case of modeling a person who might have kind of these discret beliefs with their own relative strengths um I would view it as you know this is this is a discret per time step multiple time steps per trial of the simulation um you know I view it as like there's an action perception Loop so first and it could have your starting point it doesn't matter right it's like presumably this Loop is infinitely recurring so to speak until it can't anymore because something happened with your biology to where you don't exist anymore you lost the part that's doing the inference or otherwise um but it would begin with something in an algorithmic fashion as something like an agent engages with their Environ in the sense of they sense something from the uh from the environment so they take in an observation as it were they they could see it it could be Inception and they they feel it or even notice a signal coming from within uh exteroceptive sensation would be that the seeing it you know the the more typical way of viewing it is like oh we have five senses and one of them is touch and one of them is taste and so on that that would be more exteroceptive we'd also have proprioceptive in the sense of you know physical movement which uh uh you know as far as we understand the brain I mean it that has its own distinct set of Dynamics so on um in any case you you take in some kind of um observation and then from there you update your beliefs right it's as if your mind is processing that information you know we can imagine it very much in a real- time setting and so it's there that these inference processes play out what does that water bottle that I saw mean right and then uh and then from there I can decide what to do about that fact or it doesn't have to be about that fact it might just be that I saw it but but nonetheless what am I what am I going to do right now I could choose to do nothing I continue sitting here right it a null action is still an action as it were we're always doing something um and then the cycle just kind of well final important part that that I'm no longer responsible for once I ch an action you know I do that action and then that reciprocally impacts the environment around me so if I just sit here uh you know you all will see me sitting here and uh and and there's not much going on I could jump up and dance around for no reason and upset all of my neighbors in my apartment complex I suppose that's another thing I could do which would then I would would elicit probably an observation to me of someone knocking on my door telling me to keep it down or maybe they're more polite than that but anyway but you can see the point right it's like it's you're able to once you kind of get the the notion of the action perception Loop down and get a a fair intuitive intuitive enough grasp of the concepts you can kind of start to become much more granular with it view it as a series of processes and then whenever you want to model something then it suddenly like that can give you that kind of grasp of like oh this is how I would model like this is how I would drill down into trying to understand these kinds of behaviors that I want to see such as me looking at PTSD and getting a verbal description from uh you know a uh a qualitative study uh participant who has PTSD describing how their day tends to go or how do they react in particular circumstances you know with PTSD it's like people get um triggered um and so and I I actually have it myself so there's a there's a personal interest there uh for me and so um to to actually be able to look at the behavior and then consider like what is happening there what what what is distinct about that and then if we start to view model parameters actually as you know these kind of things a generative model that we have that allows us to navigate the world around us and engage in different things then suddenly those parameter values become incredibly important because if they become too misaligned suddenly I start engaging in different kinds of suboptimal behavior or someone else would right so um they you the idea of M repeatedly making the same mistake over and over despite getting evidence showing you that you should not be doing something or you should should be doing something right it's in that case that it's kind of like we can start to view a model is something that is kind of just you know engaging in a certain behavior and it's actually the parameter values and the construction of the model that that kind of determin it's its sort of Fitness or optimality for the task at hand you know so you can imagine know someone who who who has ADHD and might have some kind of difficulty with with concentrating and and emotional regulation and so whenever they try to engage in Social settings they might struggle with engaging in the conversation in the fashion that uh they would actually like to like there's there's a difference between what we think and what we do right um so so it might be that due to particular characteristics of their neurobi neurobiology synaptic transmission and so on uh it might be kind of preventing them or kind of skewing them towards certain kinds of Behavior Uh both internally and externally that that don't actually meet meet the situation uh it's around here that I think anyone who's interested in things like actually something very a very different example which is machine learning would suddenly like you know their ears would perk up because they'd be thinking about how this is a model that is making poor predictions and so it's act behaving in a very suboptimal way relative to the sort of purpose or or or intent of the model if we could use the word intent there because I've been talking about humans so much um so very long explanation but I I hope I hope that was useful and and maybe interesting to at least some of you um and it has been fun talking in these kinds of ways with with people who are you know with the clinical therapist I you know who I'm doing the research pract like the research group um with so yeah hey Andrew yes um in this chapter you also mentioned it and they don't say much about learning um that's something I'm I am very interested in I actually I came to active inference from this paper of laa de Pao about um active inference I think it's called active inference goes goes to school about large language models uh in education um so I'm like very interested in um the notion of learning in in active inference um because it's seems to me that it's like very fundamental aspect of living organisms so maybe you can say a little bit about that or and also I don't know if uh if it's something that goes more into detail in the next chapters and I would like to a little bit about learning yeah yeah definitely no absolutely and yeah I think I think part of why they don't go too far into learning is in part uh and I I'll focus more on actually directly answering your question but I think they maybe don't they Dodge going too quickly into it in chapter one just because we're already being presented with at face value like this is a normative framework and they're you know it's going to take time to get up to there and then furthermore I think by explaining inference itself as its own process it it it it's a good introduction prior to getting into learning which is where instead of like with inference we we update our beliefs to get these in a basian framework these posterior beliefs it's like your prior beliefs of what you you thought before versus you know some kind of Engagement with the world through perception or action then you end up with what you think after but we can view learning as a very similar process perhaps it happens at a slower time scale um you in most modeling it it very much is you know you could almost think of it as the difference between short long-term memory if we put them in caricature a little bit just for the sake of example um but um so so learning would be like that it will first of all the the textbook most certainly will get much more into learning um I think there's a significant amount of it in the second half of the book um because it because of the kind of distinctness of it when ever engaging in actually building models and S simulating them with programming or otherwise however it will get into it much earlier on uh as well so so rest assured it'll it'll come up soon enough um yeah and and this is also a rather interesting looking paper I came across this at one point because I've also do some work with llms and I'll briefly mention that we've actually had talks previously with with Ben White as well as AAL who I think just did one a few days ago um okay Axel constant we know Andy Clark yeah anyway sorry and I think also La was in the yeah did had had one time with Laura um there's also this common double usage of terms from um learning attention memory and like are they being applied to the person and like the real question of human learning in a bigger form or in it's sort of narrower just like parameter updating belief updating but that's not necessarily like the experiential qualia belief um yeah this was this was yeah a great um paper and it's a good direction to keep exploring yeah yeah that's um and also the question you you put in the in the chat it's it's like a very core question of to think about how this uh framework can INF form certain practices uh in the case of education but there might be another um practical um cases where this framework might inform how to use certain Technologies in this case AI systems but so very I'm going to watch that for sure yeah most definitely I mean it's active inference is only more ever increasingly more broadly being applied to phenomena that when I first came to it personally I was not necessarily interested in myself I mean I whenever I first started I was quite new to computer programming to machine learning or anything along those I keep bringing that up because active inference it has heavy overlap with with reinforcement learning um but also yeah just just Behavioral aspects of engaging with different things right now I'm uh I've been giving feedback to someone who's writing a draft on applying this L quite directly to the idea of like human uh technological kind of intera interaction we're looking at socio Technical Systems looking at technology is a sort of potential mediator of our interactions with one another and um it's it's really fascinating and it's it's just the the kind of Concepts that come in the first part or first half of this book um you know with with spending a little bit of time with them they start to provide just sort of a lot of fodder or or or at least at least initial inspirational material for thinking about these things in this way um you know when you and of course in philosophy there are a lot of theories of things like this like extended cognition uh you know the idea that is is the fact that an imense amount of information on the internet that can then be you know transmitted and received by your phone and then phone being in your pocket therefore you have this wealth of knowledge in your pocket at any given moment like uh what does that say about us if we do agree with this idea that we we infer and learn and at some point we end and and something else begins like an agent in their environment you'll see a concept later that becomes really important in active inference there have been various discussions and even debates about it so it's worth being aware of that um what it is called a marov blanket so um I personally don't get to heavy emphasis upon the word personally um I don't get too caught up in the heavy specifics of it there are many heavy like very precise specifics about it it relates partially to Judea pearls theory of the marov blanket but but it's used a bit differently here in active inference uh but one way of viewing it is simply that it's it's a way of kind of separating an agent or organism or whatever your system of interest is from what it is not um because those who are familiar with any kind of modeling are surely aware and I was too because I did my own forms of economic modeling and kind of social science oriented modeling before I I came to active inference it's like yeah like I'm looking at data and information and it gets processed in different ways and I have to decide what something isn't in order to study it and it seems like such a simple thing and then of course like for a modeler it's like you know you you still have to make certain kinds of definitive statements over what your system is versus what is the not system as I've heard uh some researchers say like you have a system and a not system you have an agent you have an environment you have uh a generative model then you have the generative process in which that generative model kind of interacts so um yeah and so with socio Technical Systems and interaction between a human and an llm so you know it's kind of like the the hum oh yeah go ahead sorry I was gonna say that that's fascinating um uh maybe I will ask you for some references uh regarding that because um yeah as I as I said in the intro it's like I I use these kind of Frameworks as a lens to understand certain things especially regarding technology so that's why I like the this paper and also uh work by Ben White in poito um because it's like a way of understanding how we uh embodied beings kind of in interact with Technologies and how we're affected by it but also how we shape them and I think that's but also comes this point you're you're you're mentioning like where does the techn technological environment ends and we start uh and that's a good question if you are trying to think about uh even different questions on ethics and regulations right how where do you put that line and so yeah maybe I will send you an email for some references because I think this is um very uh close to my work very very interesting cool yeah no looking forward to it it's it's been a lot of fun getting more involved with the The Institute and I I notice uh folks have been asking me many questions but I I welcome them so please go for it I already here and we we're about to have to wrap up here unfortunately but I look forward to next week but I just noticed here llms could be powerful educative tools in the hands of teachers augmenting the growing skilled capabilities of children here's an example of how this might look like uh as a child is engaged in a writing task rather than making auto corrections the system could highlight mistakes draw the child's attention and suggest helpful resources for self-correct so this is me not having read the paper though I believe I've skimmed it at one point um here's an example a child is engaged in writing task there's a task there's a goal here we have the the makings of a behavioral task or experiment uh right before us rather than making autoc Corrections which would be where the llm does something that the child otherwise would do the system could highlight mistakes drawing the child's attention attention is highly implicated in active inference I think many people who do anything related to to Neuroscience you know get interested in attention at some point or another as do those who work on modeling in fact people who who build llms are concerned with things like an attention mechanism that the llm either has within it or is you know kind of has is equipped with um so so so this is proposing at face value at least but I I I feel like it's rather accurate uh is something like uh Le trying to devise a way of integrating Ln such that they augment the child's ability to actually actively learn it helps to point out where the mistake is so they can you know over time learn how to identify the mistakes themselves and then also they have to do the work to correct it themselves as well which I think so the notion is this is providing more fulfilling you know uh learning experience but then also they're they're really paying attention to the sort of terminology you know there is a task at hand there is something that needs to be corrected there are attentional mechanisms in place so and then suggest helpful resources like this is providing them with what we might call epistemic value or Information Gain or just simply this is helping them to also learn additional information in addition to just uh correcting mistake it's like oh here's more info that can help me going forward and they then they themselves have to actually click on the link or whatever happens you know it's just making the suggestion so this this prevents the LM for from becoming an additional organ on the child that they now have to always rely upon right if you want to view it in that kind of human techn technology interaction it's like instead of making this a necessary organ that you're always going to need it in order to write anything because now you're highly dependent upon autocorrect instead it's helping to augment your education ability to write by simply giving you some starting steps on getting uh used to identifying mistakes and then seeking out further resources whenever you're not sure what to do that was my little four minute of a a summary of a paper I still need to read myself but it looks really interesting so so thanks for sharing this and yeah feel free to reach out and then we also The Institute has a Discord uh and so you're you're welcome to reach me there uh it's well and then I will also ideally be around uh for next week's textbook group as well so sounds good one week today yeah thank you Andrew thank you everyone so yeah this time slot next time will be um chapter 7 discreet time which again interleaving them is totally good and and useful but if you want to continue with chat with the early part of the book then it's going to be earlier next week so and vice vers so thank you though see you all later